1. Define Embedded Systems. What are the components of embedded systems.

Embedded System :-

An embedded system is an electronic / electro-mechanical system designed to perform a specific function and is a combination of both hardware and firmware (software).

Components of Embedded Systems

The components of an embedded system can be grouped into hardware and software components.

Hardware Components :

→ Microcontroller / Microprocessor : This is the brain of Embedded system. A microcontroller (e.g. ARM Cortex, AVR) includes a CPU, memory and I/O ports in a single chip, while a microprocessor (eg : Intel, AMD) typically only includes a CPU and requires external memory and I/O components.

→ Memory :

1. RAM : Temporary storage for data being processed
2. ROM : Stores the firmware (permanent software) required to run the system.
3. Flash Memory / EEPROM : Non-volatile memory used for storing code & data, often used for updates

→ Timers / Counters : Used for managing time-dependent tasks, scheduling operations & generating interrupts.

→ Input / Output ports : Interface for communication with external devices like sensors, actuators, etc.

→ Analog-to-Digital Converters (ADC) : Converts analog signals from sensors into digital data for processing.

→ Digital-to-Analog Converters (DAC) : Converts digital output signal

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into analog form, if needed.

→ Sensors / Actuators :

Sensors : Gather data from the environment

Actuators : Execute actions based on the system's decisions (eg: relays, motors)

→ Communication Interfaces

- Serial Communication (UART, SPI, I2C) : Allows Communication between the microcontroller and peripherals.
- Wireless Communication (Wi-Fi, Bluetooth, Zigbee) : Enables wireless data transmission.

→ Power Supply : Provides necessary power to the system, often includes voltage regulators to ensure stable operation.

→ Clock / Crystal oscillator : provides timing signals to synchronize system operations.

Software Components :

→ Firmware : Low-level software programmed into the ROM that controls the hardware functions. It is typically written in languages like C or assembly.

→ RTOS (Real - Time - Operating - System) : Provides multitasking, scheduling, and time-critical execution in embedded systems.

→ Drivers : Software that allows the system to Communicate with hardware components such as I/O devices

→ Middleware : Software libraries or services that act as a bridge between the operating system and application software

→ Application software : This software performs the high-level functions specific to the task that the embedded system is designed to handle.

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2. Write short notes on history of Embedded System.

Embedded systems were in existence even before the IT revolution. In the olden days embedded systems were built around the old vaccum tube and transistor technologies.

In 1960 Embedded systems were first used for developing the Apollo Guidance System (AGS)

In 1965 Autonetics D-17B Computer for Minute Man Missile Guidance System.

Apollo Guidance System (1960):-

The Apollo Guidance System (AGS) was a pioneering use of embedded systems, developed by the MIT Instrumentation Laboratory for NASA's Apollo missions.

It was designed to control spacecraft operations, navigation, and guidance during manned lunar missions.

The AGS consisted of two main components: the Command Module's Primary Guidance, Navigation, and Control System (PGNCS) and the Lunar Module's Abort Guidance System (AGS).

Autonetics D-178 (1965)- Minute Man Missile Guidance System:-

The Autonetics D-17B was a specialized computer designed for the Minuteman I missile guidance system. The Minuteman missile was an Intercontinental Ballistic Missile (ICBM) used during the Cold war by the United States, and its success heavily relied on the accuracy of its guidance system.

The D-17B was a fully transistorized digital computer, meaning it did not rely on vacuum tubes, making it more reliable and compact.

The system was designed to handle real-time calculations necessary for guiding the missile's trajectory from launch to the target.

Later MicroControllers & Micro processors are introduced.

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3. What is the purpose of Embedded Systems Explain.

Purpose of Embedded Systems:

Embedded systems are used in various domains like consumer electronics, home automation, healthcare etc. Each embedded system is designed to serve the purpose of any one or combination of the following tasks:

1. Data Collection / Storage / Representation.
2. Data Communication.
3. Data (signal) processing
4. Monitoring
5. Control
6. Application specific user interface

1. Data Collection / Storage / Representation:

Embedded systems designed for the purpose of data collection performs acquisition of data from the external world.

Data Collection is usually done for storage, analysis, manipulation and transmission. Data refers to all kind of information, viz. text, voice, image etc. It can be analog / digital.

The collected data may be stored directly in the system or may be transmitted to some other systems or it may be processed by the system or it may be deleted instantly after giving a meaningful representation.

2. Data Communication:

Embedded data communication systems are deployed in applications ranging from complex satellite communication system to simple home networking systems.

The data collected by an embedded terminal may require transferring of the same to some other system located remotely. The transmission is achieved either by a wire-line medium or by a wire-less medium.

The data collecting embedded terminal itself can incorporate data communication units like wireless modules (Bluetooth, ZigBee, Wi-Fi, EDGE, etc). or wire-line modules (RS-232C, USB, TCP/IP etc).

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3. Data (Signal) Processing :

The data collected by embedded systems may be used for various kinds of data processing.

Embedded systems with signal processing functionalities are employed in applications demanding signal processing like speech coding, synthesis, audio video codec, transmission applications, etc.

A digital hearing aid is a typical example of an embedded system employing data processing.

4. Monitoring.

Embedded systems falling under this category are specifically designed for monitoring purpose. Almost all embedded products coming under the medical domain are the monitoring functions only.

They are used for determining the state of some variables using input sensor. Ex:- ECG, CRO etc.

5. Control :

Embedded system with control functionalities impose control over some variables according to the changes in input variables.

A system with control functionality contains both sensors and actuators

Sensors are connected to the input port for capturing the changes in environmental variable or measuring variable.

The actuators connected to the output port are controlled according to the changes in input variable to put an impact on the controlling variable to bring the controlled variable to the specific range.

Ex:- Air Conditioner (AC).

6. Application Specific User Interface

These are embedded systems with application specific user interfaces like buttons, switches, keypad, lights, bells, display units etc. Mobile phone is an example for this. In mobile phone the user interface is provided through the keypad, graphic LCD module, vibration alert etc.

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Determine the Major Application Areas of Embedded Systems.

(5)

Major Application Areas of Embedded Systems :

→ Embedded technology has acquired a new dimension from its first generation model, the Apollo guidance computer, to the latest radio navigation system combined with in-car entertainment technology and microprocessor based smart appliances.

→ The application areas and the products in the embedded domain are countless

→ A few of the important domains and products are listed below:

1. Consumer electronics : Camcorders, Cameras etc.
2. Household appliances : Television, DVD players, washing machine, fridge, microwave oven, etc.
3. Home automation and security systems : Air Conditioners, Sprinklers, intruder detection alarms, closed circuit television cameras, fire alarms, etc.
4. Automotive industry : Anti-lock breaking system (ABS), engine control, ignition systems, automatic navigation systems, etc.
5. Telecom : Cellular telephones, telephone switches, handset multimedia applications, etc.
6. Computer peripherals : Printers, Scanners, fax machines, etc.
7. Computer networking systems : Network routers, switches, hubs, firewalls, etc.
8. Healthcare : Different kinds of Scanners, EEG, ECG machines etc.
9. Measurement & Instrumentation : Digital multi media meters, digital CRO's, logic analyzers PLC systems, etc.
10. Banking & Retail : Automatic teller machines (ATM) and

currency counters, point of sales (POS)

11. Card Readers: Barcode, smart card readers, hand held devices etc.

(6) Explain the Core of the Embedded System.

Core of Embedded System

Embedded systems are designed for specific applications, and their core components can be categorized into the following types:

1. General purpose and Domain specific processors
   → Microprocessors
   → MicroControllers
   → Digital Signal Processors
2. Application Specific Integrated Circuits (ASICs)
3. Programmable Logic Devices (PLDs).
4. Commercial off-the-shelf Components (COTS)

1. General purpose and Domain Specific processors

Almost 80% of the embedded systems are processor/controller based. The processor may be a micro-processor or a microcontroller or a digital signal processor, depending on the domain and application.

Microprocessors: Versatile processors designed for general computing tasks, capable of running various applications. Commonly used in computers and some embedded applications.

A Microprocessor is a silicon chip representing a central processing unit, which is capable of performing arithmetic as well as logical operations according to a pre-defined set of instructions.

MicroControllers: A microcontroller is a highly integrated chip that contains a CPU, scratch pad RAM, special and general purpose register arrays, on chip ROM/FLASH memory for program

storage, timer and interrupt control units and dedicated I/O ports. Microcontrollers can be considered as a super set of microprocessors.

They found greater place in the embedded domain in place of microprocessors. Apart from this, they are cheap, cost effective and readily available in the market.

Digital signal processors : DSP's are powerful special purpose 8/16/32 bit microprocessors designed specifically to meet the computational demands and power constraints of today's embededded audio, video and communication applications.

DSP's implement algorithms in hardware which speeds up the execution.

Application Specific Integrated Circuits (ASICs):

ASIC is a microchip designed to perform a specific or unique application. It is used as replacement to conventional general purpose logic chips.

It integrates several functions into a single chip and there by reduces the system development cost. Most of the ASICs are proprietary products. As a single chip, ASIC consumes a very small area in the total system and thereby helps in the design of smaller systems with high capabilities/ functionalities.

Programmable Logic Devices:

Logic devices provide specific functions, including device-to-device interfacing, data communication, signal processing, data display, timing and control operations and almost every other function a system must perform.

Logic devices can be classified into two broad categories - fixed and programmable. With programmable logic devices, designers use inexpensive software tools to quickly develop, simulate and test their design.

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Commercial Off-the-shelf Components (COTS)

A commercial Off-the-shelf (COTS) product is one which is one which is used 'as-is'. COTS products are designed in such a way to provide easy integration and interoperability with existing system components.

8. Explain the Characteristics of an Embedded System.

(8) The important characteristics of an embedded system are as follows.

1. Application and Domain specific
2. Reactive and Real Time
3. Operates in harsh environment
4. Distributed
5. Small size and weight
6. Power Concerns.

Application and Domain Specific :-

Embedded systems are designed to perform a specific task / function within a particular domain. Unlike general-purpose computers, which can run various applications, embedded systems are optimized for a specific application.

Ex:- washing machines, Micro wave etc

Reactive and Real-Time :

Embedded systems often need to react to inputs from their environment in real time. They continuously monitor and respond to stimuli, such as sensor inputs, within strict timing constraints.

Ex:- anti-lock braking system (ABS) in cars.

Operates in harsh environment :-

Many embedded systems are designed to operate in extreme conditions such as high / low temperatures, humidity or vibrations. For instance, embedded systems in industrial machinery / automotive systems must be robust enough to work in ~~duty~~ dusty, noisy or high-temperature

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environments without failing.

Distributed :-

Embedded systems are often part of a larger distributed system, where multiple embedded systems work together to achieve a common goal.

For example, in modern cars, various embedded systems communicate with each other to ensure smooth operation.

Small size and weight

Space and weight are critical constraints in embedded systems, especially for portable / handled devices. These systems need to be small and lightweight to be integrated into the product without adding unnecessary bulk.

Ex:- Embedded systems in mobile phones, wearable devices or medical implants.

Power Concerns :-

Since many embedded systems run on battery power or have limited energy sources, they need to be energy efficient. Power optimization is critical, especially in portable devices like smartphones, sensors, or medical equipment, to extend battery life and ensure reliable performance with minimal energy consumption.

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4. Classification of the Embedded System./Various types of embedded systems.

Classification of Embedded Systems:

Embedded systems can be classified based on various criteria. Below are four key categories used for classification.

1. Based on Generation:

- First Generation: These embedded systems were built using simple, 8-bit or 4-bit microprocessors or microcontrollers. Examples include simple home appliances like washing machines/calculators.
- Second Generation: With advancements, 16-bit microcontrollers were used. Systems had better functionality and processing speed. Examples include early mobile phones and gaming consoles
- Third Generation: These systems employ 32-bit microcontrollers and have networking capabilities. They are used in more complex applications like modern smartphones and advanced industrial systems.
- Fourth Generation: The latest embedded systems use 64-bit processors, with improved real-time performance, greater complexity, and IOT examples include smart wearables and autonomous vehicles.

~~Based on~~
2. Based on complexity and performance Requirements:

- Small scale embedded systems: These are simple, low-performance systems with limited computational power, typically built using 8 bit or 16 bit microcontrollers.
  Ex:- toys, electronic thermometers and simple sensors.
- Medium-scale embedded systems: They are more complex and use 16bit or 32 bit microcontrollers. These systems are designed for specific applications with higher processing power, such as automotive controls, medical devices, and industrial machines.
- Large-scale Embedded Systems: These are highly complex systems that require powerful processors and vast memory.

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They are used in advanced applications like robotics, and modern smartphones, where performance and real-time processing are critical.

3. Based on Deterministic Behavior:

- Hard Real-Time Systems: These systems have strict timing constraints, where a failure to meet deadlines can result in system failure Ex:- air bag control systems, and flight control systems.
- Soft Real-Time Systems: Timing is important, but not critical. Even if deadlines are missed, the system continues to function, though performance may degrade. Ex:- multimedia systems and video streaming applications.

4. Based on Triggering

- Event-Triggered Embedded Systems:
  These systems perform tasks when specific events or conditions occur. An external or internal event triggers the system's response. Ex: security alarms and smart home systems.
- Time-Triggered Embedded Systems:
  These systems execute tasks at predefined time intervals, regardless of external events. They are driven by internal clocks or timers. Ex:- periodic data collection systems, time-driven control systems and scheduled communication systems.

(11)

5. Briefly explain about memory concept in embedded system.

Memory :-

Memory is a crucial component of embedded systems, as it stores the software and data required for system operation.

Embedded systems typically have limited memory resources, and memory design is optimized to meet real-time and application-specific constraints.

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Memory in Embedded System can be broadly classified into 'program Memory' and 'Data Memory'

Program Memory :

Program memory is used to store the code or firmware that the embedded systems runs. This is typically non-volatile memory, meaning the data is retained even when the power is switched off.

- ROM (Read Only Memory)
  → stores the embedded system's firmware or software permanently.
  → Cannot be altered during normal operation.
  → used for storing critical system instructions that do not change.

Types of ROM:

- PROM (Programmable ROM): Can be programmed once after manufacturing. Once programmed, it cannot be modified or erased.
- EPROM (Erasable Programmable ROM): Can be erased and reprogrammed by exposing it to ultraviolet (UV) light, allowing updates or corrections to the firmware.
- EEPROM (Electrically Erasable Programmable ROM): Can be erased and reprogrammed electrically without needing to remove the chip from the system. It allows for more frequent updates ~~from~~ compared to EPROM.
- Flash Memory : A modern type of EEPROM, flash memory allows fast read and write operations and can be reprogrammed in blocks. It is widely used in embedded systems for storing firmware and other non-volatile data.
- MROM (Masked ROM): It is pre-programmed during the manufacturing process. It is an early, basic form of ROM and plays a role in the memory structure of embedded systems, especially in cost-sensitive or high volume production environments.

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Data Memory:

Data Memory is used to store temporary data required during the system's operation. Unlike program memory, it is typically volatile, meaning the data is lost when the system is powered off.

- RAM (Random Access Memory):

RAM stores variables, temporary data, and input/output data during program execution. It is essential for dynamic data handling during real-time processing.

It is volatile memory.

Types

- SRAM (Static RAM):

Faster but more expensive than other types of RAM. It does not require refreshing to maintain the data, making it useful for applications needing high-speed memory access.

Often used in cache memory for small, high-speed applications within ES.

- DRAM (Dynamic RAM)

Slower and less expensive than SRAM

Requires periodic refreshing of its data

Used when larger memory capacity is needed at a lower cost, typically in more complex systems.

- Non-Volatile RAM:

NVRAM: Combines the features of RAM with non-volatile characteristics, meaning data is retained even when power is turned off. It is useful in applications that need to retain data between reboots, such as in configuration data or system logs.

FRAM (Ferroelectric RAM): A type of non-volatile RAM that is faster and offers high endurance. It is used in specific embedded systems requiring frequent data writing and long-term retention.

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7.a. Define Sensors and Actuators in Embedded System.

Sensors :- (7a, not there for mid)

Sensors are devices that detect and measure physical quantities from the environment, such as temperature, light, pressure or motion and convert them into signals that can be read by a computer or a machine. Sensors provide data to a system, to help it understand the surroundings and make decisions accordingly.

Ex :- temperature sensors, motion sensors and pressure sensors.

Actuators :

Actuators are devices that take signals from a controller (typically electrical signals) and convert them into physical actions like movement, force or a change in the system's state.

In other words Actuators is a form of transducer device (mechanical or electrical) which converts signals to corresponding physical action (motion). Actuator acts as an output device.

They are responsible for executing the commands of a control system. Ex :- electric motors, hydraulic pistons and solenoids.

(9) 8. Explain the processor and OS trends in Embedded system.

Processor trends in Embedded System :

Processor trends in embedded systems focus on improving performance, power efficiency, flexibility, and security to meet the growing demands of modern applications.

The 3 key trends are

1. System on Chip (SoC)
2. Multicore Processors
3. Reconfigurable processors.

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System on chip (SoC):

SoC integrates various components, such as the CPU, memory and peripherals, onto a single chip, minimizing the size and power consumption. SoCs are used in devices like smartphones and IoT devices due to their compact design and efficiency.

Multicore processors:

Multicore processors contain multiple cores, allowing parallel processing of tasks. This improves system performance and efficiency, especially in real-time and multitasking applications such as automotive systems and industrial automation

Reconfigurable Processors:

Reconfigurable processors, like those using FPGAs, can be customized and reprogrammed post-manufacture for specific tasks. This allows for flexibility and optimization in applications like telecommunications and AI, where hardware needs to adapt to changing requirements.

Embedded OS Trends :-

Embedded OS trends refer to the evolving practices in the embedded operating system industry, driven by advancements in processor technologies and the increasing need for efficiency, flexibility, and customization in embedded devices.

Microkernel Architecture:

Modern embedded OSs are transitioning to a microkernel architecture, where the kernel contains only essential services, while additional features operate in user space. This enhances modularity, security and flexibility compared to monolithic architectures.

Device-Oriented Design:

OS suppliers now offer off-the-shelf OSs tailored for specific device categories, minimizing customization efforts. For instance, microsoft provides embedded OS versions for point-of-sale terminals, media players, and set-top boxes.

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Custom OS for Devices:

Companies like microsoft develop specialized OS. versions for specific device families, such as windows mobile for mobile handsets.

Growth of open-source Embedded OS:

The popularity of open-source OSs, particularly those based on Embedded Linux, is rising offering customizable and cost-effective solutions for diverse embedded applications, supported by community contributions.

Support for Multicore processors:

With the shift from single-core to multicore processors, embedded OSs are adapting to support these architectures. Ex:- the ~~lat~~ latest versions of VxWorks RTOS are designed to handle multicore processing efficiently.

7. What is sensor Explain its role in embedded system design illustrate with example.

Sensor:-

A sensor is a device that detects and measures a physical property from the environment (such as temperature, light, pressure, or motion) and converts it into a signal that can be read by an observer or by an instrument, like a microcontroller or processor.

Role of Sensors in Embedded System Design:

Sensors are integral to embedded systems because they provide real-time data inputs from the physical world. The embedded system processes this input to make decisions, control devices or provide feedback. They play a key role in:

1. Data Collection:

Sensors gather critical information, which is processed by the system for decision-making.

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2. Automation:

In applications like home automation, healthcare, and industrial systems, sensors enable automatic responses to changes in the environment.

3. Monitoring and Control:

Embedded systems use sensor data to monitor conditions and trigger specific actions (e.g: adjusting temperature, turning on lights).

4. Feedback Loop:

In control systems, the sensor's data can provide feedback to maintain desired system behavior. (e.g adjusting speed in motors).

Example:-

Temperature sensor in a smart Thermostat.

In a smart thermostat system, the temperature sensor continuously monitors the room temperature. This data is sent to a microcontroller, which processes the information and compares it with the user defined temperature settings.

Based on this comparision, the embedded system makes decisions:

→ if the room is colder than the set temperature, the system might turn on the heating.

→ if it is hotter, it might activate the air conditioning.

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12. What is actuator Explain its role in embedded system design Illustrate with example.

Actuator :-

An actuator is a device that converts electrical signals from a control system into physical motion or action. It receives signals from the embedded system, often based on sensor inputs, and performs actions such as moving, rotating, opening or adjusting something in the physical world.

Role of actuators in embedded System Design:

In embedded systems, actuators are the output devices that enable the system to interact with its environment. While sensor collect data from the environment, actuators respond to processed sensor data and trigger real-world actions.

Their key roles include:

Physical Control: Actuators physically control devices (motors, valves etc). based on decisions made by the embedded system.

Automation: They enable automatic control and movement in systems such as robotics, home automation or industrial machinery.

Feedback loop Execution: In a feedback loop, actuators adjust system behavior (eg, speed, direction, force) to meet desired outcomes.

Energy conversion: Actuators convert electrical energy into mechanical energy or other forms of energy (like heat or light).

Types of Actuators:

→ Electrical actuators: Converts electrical energy into motion (e.g, motors)

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→ Hydraulic Actuators: Use fluid pressure to move objects
→ pneumatic Actuators: use compressed air to create movement
→ Thermal Actuators: Convert heat into motion (e.g. thermoelectric actuators)
→ Magnetic Actuators: use magnetic fields to create motion.

Ex:- Actuator in an Automated Door System.

In an automated sliding door system, actuators are used to open and close the door.
working as follows.

Sensor input: A proximity sensor detects a person near the door and sends the signal to the microcontroller.

Micro Controller decision: It processes the input and sends a command to the actuator to open the door.

Actuator action: The electric motor / hydraulic actuator receives the signal and moves the door, allowing it to open.

Feedback: After the person passes through, the system may send a signal for the door to close.

10. Discuss Quality attributes of Embedded System.

Quality attributes of Embedded System:-

Quality attributes are the non-functional requirements that need to be documented properly in any system design.

If the quality attributes are more concrete and measurable it will give a positive impact on the system development process and the end product.

The various quality attributes that needs to be addressed in any embedded system development are broadly classified into two, namely 'Operational' and 'Non-operational'

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quality attributes.

Operational Quality Attributes:

The operational quality attributes represent the relevant quality attributes related to the embedded system when it is in the operational mode or 'online' mode. Attributes as follows.

Response: Measures how quickly the system reacts to inputs. Crucial for real-time applications where fast actions are needed (e.g automotive braking).

Throughput: Refers to the system's ability to process data or tasks efficiently over time. Important for high-load applications like routers and multimedia systems.

Reliability: Indicates how consistently the system performs without failures. Essential in critical systems like medical devices or aerospace systems.

Maintainability: Defines how easily the system can be repaired, updated, or modified. Important for reducing downtime and costs in long-term use.

Security: Ensures the system is protected from unauthorized access and cyber threats. Vital for connected devices like IoT systems.

Safety: Prevents the system from causing harm or damage in failure conditions. Critical in areas like automotive and healthcare where safety is paramount.

Non-Operational Quality Attributes:

The quality attributes that needs to be addressed for the product 'not' on the basis of operational aspects are grouped under this.

Attributes as follows.

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1. Testing & Debug-ability:

→ Testability: Ease of verifying the system meets requirements through testing.

→ Debug ability: Simplicity in diagnosing and fixing issues.

→ Reduces the development time and enhances quality assurance.

2. Evolvability:

→ Ability to adapt to changes and integrate new features without significant redesign.

→ keeps the system relevant as technology evolves, minimizing the need for replacement.

3. Portability:

→ Ease of transferring the system across different environment or platforms

→ Extends market reach & reduces adaption costs.

4. Time to prototyne and Market:

→ Measures the time taken to develop a prototyne and launch the product

→ Shorter cycles enable quicker feedback and faster market delivery.

5. Per unit and Total Cost:

→ Assesses the cost of production per unit and overall lifecycle costs.

→ Helps in competitive and maximizing profitability.

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